

MICROBIOLOGY – SHORT EXAM NOTES

Preparation of Culture Media – Solid and Liquid (Ananthanarayan & Paniker, Ch. 3–5 + standard practical)

The textbook describes the ingredients, types and sterilisation of media but not the bench steps of preparation; the step-by-step procedures, quantities and QC below are standard practical content.

1. INTRODUCTION AND INGREDIENTS

A **culture medium** is a mixture of nutrients that supports the growth of microorganisms and must provide a source of **carbon and energy**, nitrogen, salts and water. **Liquid media (broths)** give diffuse growth; **solid media** are made by adding **agar** (sometimes heated serum or egg) and give **colonies**.

Common ingredients (from the book)

Ingredient	Nature / role
Water	Distilled/deionised; vehicle for nutrients
Peptone	Partially digested protein – proteoses, polypeptides, amino acids, phosphates, K, Mg, growth factors (riboflavin); nitrogen source
Meat / beef extract, yeast extract	Amino acids, vitamins, minerals
Sodium chloride	Maintains osmotic balance
Agar	From seaweed; no nutritive value , not affected by bacterial growth; melts at 98°C, sets at ~42°C; 2% for solid, ~0.5% for semisolid media
Blood, serum, egg	Enrichment for fastidious organisms (blood agar, Loeffler's serum, L-J medium)
Sugars, indicators	Fermentation tests – phenol red, neutral red, Andrade's indicator
Selective agents	Bile salts, dyes (malachite green), antibiotics (vancomycin, colistin, nystatin), tellurite

Media are prepared either from **individual ingredients** or, commonly today, from **commercial dehydrated powders** reconstituted as per the manufacturer's instructions (grams per litre printed on the container).

2. GENERAL STEPS IN MEDIA PREPARATION

1. **Cleaning of glassware** – flasks, tubes, Petri dishes washed, dried and sterilised in a **hot-air oven (160°C, 2 h)**.
2. **Weighing** – weigh the ingredients / dehydrated powder accurately on a balance.
3. **Dissolving** – add to the required volume of **distilled water** in a flask; mix; **heat to boil** (with stirring) for agar-containing media until completely dissolved.
4. **Adjusting pH** – most media **pH 7.2–7.6** (checked with a pH meter or indicator papers; adjusted with **N/10 NaOH or HCl**). Dehydrated media are usually pre-adjusted.
5. **Dispensing** – into flasks, tubes or bottles in required volumes; plug with cotton wool or screw caps.
6. **Sterilisation** – by the appropriate method (see table below).
7. **Addition of supplements** – heat-labile additives (blood, serum, antibiotics, some sugars) are added **aseptically after cooling to 45–50°C**.
8. **Pouring / setting** – plates, slopes or deeps (for solid media).
9. **Sterility check** – incubate a sample (or whole batch) at **37°C overnight**; discard any with growth.

10. **Labelling and storage** – name, date, batch; store at **2–8°C (refrigerator)**, plates inverted in sealed bags; use within the shelf-life.

Sterilisation of Media (from the book)

Medium	Method	Condition
Most media (nutrient broth/agar, peptone water, MacConkey)	Autoclave	121°C, 15 lb/in², 15 min
Media containing sugars or gelatin	Tyndallisation (Koch/Arnold steamer) – or low-pressure autoclaving	100°C, 20 min on 3 successive days
Serum and egg media – Loeffler's serum slope, L-J medium	Inspissation	80–85°C, ½ h on 3 successive days
Heat-labile solutions (serum, sugar and antibiotic solutions, urea)	Filtration	Membrane filter 0.22 µm, added aseptically
Some selective media (e.g. TCBS, DCA, SS agar)	Boiling only (not autoclaved)	As per manufacturer

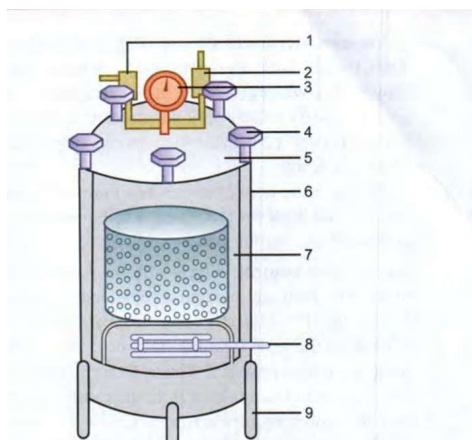


Fig. 3.3 A simple autoclave – used to sterilise most media (from the book)

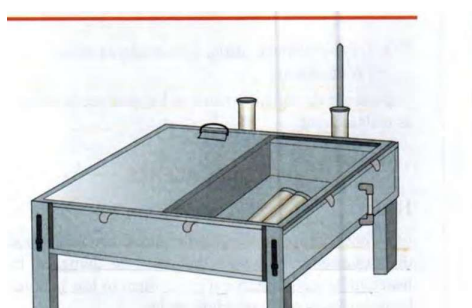


Fig. 3.2 Inspissator – for L-J and Loeffler's serum media (from the book)

3. PREPARATION OF LIQUID MEDIA

Nutrient Broth (basal liquid medium)

Ingredient	Quantity (per litre)
Peptone	10 g
Meat (beef) extract	10 g (3–10 g in different formulas)
Sodium chloride	5 g

Ingredient	Quantity (per litre)
Distilled water	1000 ml

Procedure: dissolve ingredients in water with gentle heat → adjust **pH 7.4** → filter if cloudy → dispense **5–10 ml** into test tubes or larger volumes in flasks → plug/cap → **autoclave at 121°C for 15 min** → cool → sterility check → store. Growth appears as **turbidity**, deposit or pellicle.

Peptone Water

Peptone 10 g + NaCl 5 g + water 1 L, pH 7.2–7.4, autoclaved. Used for **indole test** and as the base for **sugar media**. **Alkaline peptone water (pH 8.6)** is used as an enrichment/transport medium for *V. cholerae*.

Sugar (Carbohydrate Fermentation) Media

1% sugar in peptone water + indicator (e.g. **Andrade's**). Dispensed in tubes containing an **inverted Durham's tube** (to detect gas); sterilised by **tyndallisation** or autoclaving at lower pressure (or sugar solution filter-sterilised and added to autoclaved base). Acid production turns the medium **pink/red**; gas collects in the Durham's tube. **Hiss' serum sugars** (with 3% serum) for fastidious organisms.

Other Liquid Media

- **Brain–heart infusion (BHI) broth** – rich medium for **blood culture** and fastidious organisms.
- **Enrichment broths** – **selenite F** and **tetrathionate broth** (feces; *Salmonella*); selenite F is **not autoclaved** (steamed only).
- **Thioglycollate broth** and **Robertson's cooked meat medium** – anaerobic; **boiled before use** to expel oxygen; RCM tubes contain meat pieces covered with broth, layered with sterile liquid paraffin if needed.
- **Glucose phosphate broth** – MR and VP tests; **Koser's citrate** – citrate utilisation.

4. PREPARATION OF SOLID MEDIA

Nutrient Agar (basal solid medium)

Nutrient broth + 2% agar (20 g agar per litre). **Procedure:** add agar to broth → **boil until agar dissolves completely** (clear solution) → adjust **pH 7.4** → dispense in flasks (for plates) or tubes (for slopes/deeps) → autoclave **121°C, 15 min** → cool to **45–50°C** in a water bath before pouring.

Pouring Plates

1. Work near a Bunsen flame or in a **laminar air flow cabinet**; clean the bench with disinfectant.
2. Cool molten medium to **45–50°C** (hot enough to stay liquid, cool enough to hold and not to cause heavy condensation or kill added blood).
3. **Flame the neck** of the flask; lift the Petri dish lid slightly and pour about **15–20 ml per 90 mm plate** (depth ~4 mm).
4. Replace the lid; swirl gently to spread evenly; prick or flame away **air bubbles**.
5. Allow to **set on a level surface**; then **dry** (lids ajar in incubator/LAF) and store **inverted** to prevent condensation dripping on the surface.

Slopes (Slants), Butt-and-Slant and Deeps

- **Slope/slant** – tubes with 5–7 ml molten medium are allowed to set in a **slanting position** (e.g. nutrient agar slope, Christensen's urease).
- **Butt and slant** – set at a slant leaving a deep butt (e.g. **TSI agar**).
- **Deep / stab tubes** – set upright (e.g. for stab culture, semisolid **0.5% agar** motility medium, Craigie's tube).
- **Serum and egg slopes** (Loeffler's, **L–J**) – dispensed in slanted bottles and **coagulated/sterilised by inspissation**.

Enriched and Special Solid Media

Medium	Preparation points
Blood agar	Nutrient agar (or blood agar base) autoclaved, cooled to 45–50°C , then 5–10% sterile defibrinated sheep blood added aseptically, mixed without bubbles and poured
Chocolate agar	Blood added to base and heated to 70–80°C (about 10–15 min) until chocolate-brown – releases X (haemin) and V (NAD) factors
MacConkey agar	Dehydrated powder (peptone, lactose, bile salt, neutral red , agar) boiled to dissolve and autoclaved; poured plates are pink-red
Löwenstein–Jensen	Mineral salt solution + glycerol + malachite green mixed with homogenised whole eggs , dispensed in bottles, slanted and inspissated
Loeffler's serum slope	Serum + nutrient broth with glucose; slanted and inspissated
Selective media (DCA, TCBS, SS agar, Thayer–Martin)	Many are boiled only, not autoclaved ; antibiotics (e.g. VCN supplement) added after cooling
Mueller–Hinton agar	For antibiotic sensitivity; poured to a uniform 4 mm depth
Sabouraud dextrose agar	Low pH (~5.6); ± chloramphenicol/gentamicin added after cooling

5. QUALITY CONTROL AND STORAGE

- **Sterility test** – incubate representative plates/tubes (~5% of batch) at 37°C for 24–48 h; no growth should occur.
- **Performance test** – inoculate known **control strains** (e.g. *E. coli* ATCC 25922, *S. aureus* ATCC 25923, *P. aeruginosa*) to check growth and expected reactions (e.g. pink colonies on MacConkey, haemolysis on blood agar).
- **Physical checks** – correct pH, colour, clarity, depth, absence of bubbles, cracks or contamination.
- **Storage** – plates inverted in sealed bags at **2–8°C**; broths in screw-capped tubes; label with name, batch and date; warm to room temperature before use; discard if dried, contaminated or expired.
- **Records** – media register with batch details, sterilisation and QC results.

6. DEMONSTRATION (PRACTICAL SPOTTERS)

Spotter	Identify & write
Nutrient broth tube	Basal liquid medium – peptone, meat extract, NaCl, water; pH 7.4; autoclaved; shows turbidity

Spotter	Identify & write
Peptone water / sugar tube with Durham's tube	Indole / sugar fermentation; acid = pink, gas in Durham's tube
Nutrient agar plate / slope	Nutrient broth + 2% agar; basal solid medium; colony morphology, pure cultures
Blood agar	Enriched & differential; 5–10% sheep blood at 45–50°C; haemolysis
Chocolate agar	Heated blood agar (70–80°C); X and V factors; <i>H. influenzae</i> , <i>Neisseria</i>
MacConkey agar	Differential/indicator; lactose + neutral red + bile salt; LF pink, NLF pale
L–J medium	Egg-based, malachite green, inspissated; mycobacteria
Robertson's cooked meat	Anaerobic liquid medium; meat pieces; boil before use
Semisolid agar / Craigie's tube	0.5% agar; motility

★ EXAM FOCUS

- Viva/practical: steps of media preparation; composition of **nutrient broth and nutrient agar**; why cool agar to **45–50°C** before pouring or adding blood; difference between **blood and chocolate agar** preparation; sterilisation of **sugar media (tyndallisation)** and **L–J medium (inspissation)**; why plates are stored **inverted**.
- Key numbers: agar 2% (solid), 0.5% (semisolid); melts 98°C, sets 42°C; pH 7.2–7.6; autoclave 121°C/15 min; 15–20 ml per plate; storage 2–8°C.